

Secondary Memory Interface

This document contains information relevant to the SMI hardware.

The SMI is intended for use with devices that might otherwise slow down or interfere with system operations if they were connected directly to the system bus.

SMI supports:

8, 9, 16, 18-bit register based devices

Both RD/WR and Enable/Direction style devices (Mode80 & Mode68)

MIPI DBI and Nokia MeSSI standard devices

Use of small and large paged multiplexed NAND devices

Minimises performance impact of connecting slow devices to the main system bus

Precise SMI bus timings

SMI Architecture

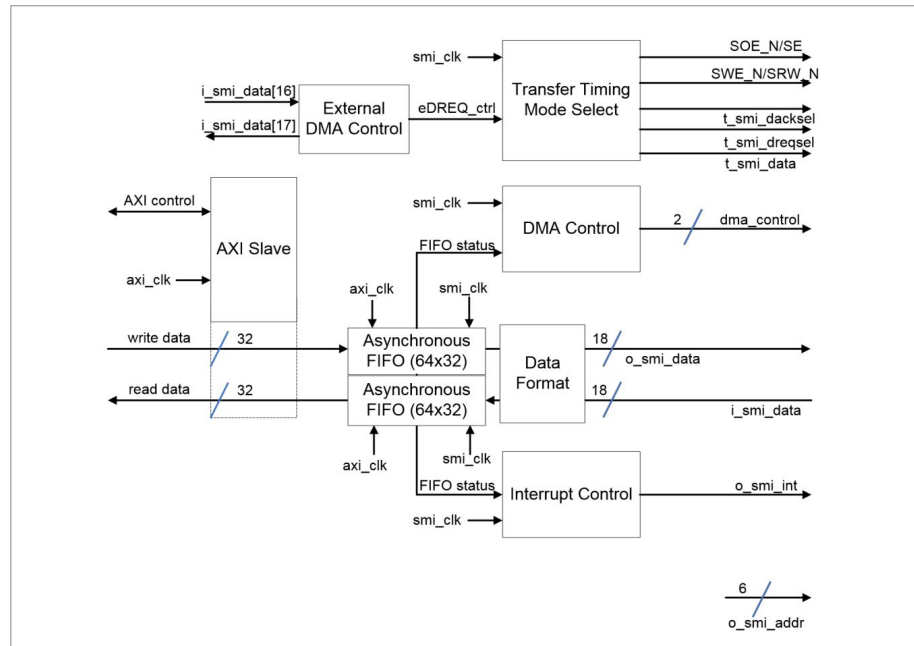


Figure 1: SMI Block Diagram

The SMI hardware is made up of three state machines: output timing to the external device, interfacing to the AXI bus and external DREQ handling. The

state machine surround two 256-byte asynchronous FIFOs, split into 64x32-bit word entries, handling read and write transfers to the device.

When writing to an external device data is written to the write FIFO from the AXI slave, clocked by the AXI clock, and read out by the device, clocked by the SMI clock.

When reading from an external device the device writes to the read FIFO, clocked by the SMI clock, and the AXI slave reads out data, clocked by the AXI clock.

There are also additional blocks to control interrupts and the DMA. External DMA requests may be passed through the SMI using the top two bits of the data bus.

Output Data Modes

The SMI can interface with 8, 9, 16 and 18-bit register devices. Data may be written to the FIFO in 16-bit RGB565 format or 32-bit XRGB8888 format. To enable pixel modes, PXLDAT must be set to 1. The format used is specified by WFORMAT in the relevant SMI_DSW register. A values bit ordering can be reversed using WSWAP. Data must be unpacked after reading from the RX FIFO with the required reads being equal to the number of FIFO entries for a given interface width and pixel format. The SMI_L register always relates to the number of physical transfers. Example: A transfer of 32-bits total volume using an 8-bit interface and RGB565 pixel format would require the SMI_L value to be 4 and the RX FIFO to be read once.

8-bit Interface

Register Map

Offset	Register Name	Description	Size
0x00	SMI_CS	SMI Control and Status	32
0x04	SMI_L	SMI Length	32
0x08	SMI_A	SMI Address	32
0x0C	SMI_D	SMI Data	32
0x10	SMI_DSR0	SMI Device Read Settings 0	32
0x14	SMI_DSRW0	SMI Device Write Settings 0	32
0x18	SMI_DSR1	SMI Device Read Settings 1	32
0x1C	SMI_DSRW1	SMI Device Write Settings 1	32
0x20	SMI_DSR2	SMI Device Read Settings 2	32
0x24	SMI_DSRW2	SMI Device Write Settings 2	32
0x28	SMI_DSR3	SMI Device Read Settings 3	32
0x2C	SMI_DSRW3	SMI Device Write Settings 3	32
0x30	SMI_DC	SMI DMA Control	32
0x34	SMI_DCS	SMI Direct Control and Status	32

Offset	Register Name	Description	Size
0x38	SMI_DA	SMI Direct Address	32
0x3C	SMI_DD	SMI Direct Data	32
0x40	SMI_FD	SMI FIFO Debug	32

Control & Status Register The status and control register controls non-direct transfers and monitors peripheral status.

Field	Description	Access
ENABLE	Enable the SMI. Value is OR'd with the DCS register. When disabled, the SMI enters low power mode.	R/W
DONE	Current transfer status. Set when the final word has been recieved or written for a given transfer	R/W
ACTIVE	Indicates the current transfer status	R
START	Start Transfer. Auto clearing bit and will always be read as 0, writing 1 while active has no effect	W
CLEAR	Clear the FIFOs. FIFOs will be reset to the empty state. Auto clearing bit and will always be read as 0	R/W
WRITE	Set the transfer direction. 0 = Read ; 1 = Write	R/W
PAD	Padding Words. The number of FIFO words to be discarded at the start of a transfer. For writes, value	R/W
TEEN	Tear Effect Mode. Programmed transfers will wait for a Tear Effect trigger before starting	R/W
INTD	Interrupt while DONE = 1	R/W
INTT	Interrupt while TXW = 1	R/W
INTR	Interrupt while RXR = 1	R/W
PVMO	Pixel Value Mode. Transmit data is taken from the pixel value interface rather than the AXI input.	R/W
SETERR	Setup error has ocured. Setup registers were written to when enabled. Can be cleared by writing 1	R/W
PXLD	Pixel Data. Enable pixel formatting modes. Data in the FIFO's will be packed to match the pixel format selected	R/W
EDREQ	External DREQ received. Indicates the status of the external devices DREQ when in DMAP mode	R
PRDY	Appear not ready on the AXI bus if the appropriate FIFO is not ready. SMI will stall the AXI bus when	R
AFERR	AXI FIFO error has ocured. Either a read of the RFIFO when empty or a write of the WFIFO when full	R
TXW	TX FIFO need writing. 0 = TX FIFO is at least 1/4 full or the transfer direction is set to READ. 1 = TX	R
RXR	RX FIFO needs reading. 0 = RX FIFO is less than 3/4 full or when DONE and FIFO no empty. 1 = RX	R
RXD	TX FIFO can accept data. 0 = TX FIFO is cannot accept new data or the transfer direction is set to READ	R
RXD	RX FIFO contains data. 0 = RX FIFO contains no data or the transfer direction is set to WRITE. 1 =	R
TXE		R
RXF	RX transfer has not filled length contract.	R

SMI Transfer length Register The SMI length register is used to specify the number of transfers on the SMI bus in words. The word width is configured by the DSR & DSW registers. During a transfer the register reads as the number of words transferred so far.

Field	Width	Description	Access	Reset
Length	31:0	Write: Sets the number of words to transfer over the external bus. Read: During a transfer contains the number of words transferred so far	R/W	0x00000000

SMI Address Register Definition The SMI address register is used to specify the address for a given transfer. Device settings can be selected using this register.

Field	Width	Description	Access	Reset
<u>X</u> DEVICE	31:10	Which set of device settings should be used for this transfer (0-4)	R/W	0x0
<u>X</u> ADDR	7:6	Address to be used for transfers and presented on the SMI bus	R/W	0x00

SMI Data Register The SMI data register is used to interface with the transmit FIFOs. Data written to the SMI_D register is placed in the transmit FIFO. Data read from the SMI_D register is taken from the receive FIFO.

Field	Width	Description	Access	Reset
DATA	31:0	Reading returns the top value of RX FIFO. Writing writes to back of TX FIFO.	R/W	0x00000000

SMI DMA Control Register The SMI DMA control register is used to control the behaviour of the DMA DREQ and PANIC signals on the AXI bus. The SMI can generate TX and RX DREQ to control an external AXI DMA controller. It can also generate PANIC signals on user configurable FIFO level threshold values.

Field	Width	Description	Access	Reset
<u>X</u> DMAEN	31:29	DMA Enable. Enables the DREQ and PANIC signals to control the AXI bus DMA transfers. 0 = No DREQ or PANIC will be issued ; 1 = DREQ and PANIC will be generated when the FIFO levels reach the programmed levels	R/W	0
<u>X</u>	27:25			

Field	Width	Description	Access	Reset
DMAP	2	External DREQ Mode. Top 2 bits of the SMI data bus are used as DREQ and DREQ_ACK signals that can be used to pace the flow of data on the external SMI bus. Separate from normal AXI DMA behaviour. Must be used in conjunction with RDREQ or WRREQ. 0 = Top two pins used as SMI data ; 1 = Top two data pins used for external DMA requests	R/W	0
PANICRX	30	RX Panic Threshold Level. An RX PANIC will be generated when the RX FIFO exceeds this threshold level. Instructs the AXI RX DMA to increase the priority of its bus requests.	R/W	0
PANICTX	10	TX Panic Threshold Level. A TX PANIC will be generated when the TX FIFO falls below this threshold level. Instructs the AXI TX DMA to increase the priority of its bus reequets.	R/W	0
REQR6	20	RX DREQ Threshold Level. An RX DREQ will be generated when the RX FIFO exceeds this threshold level. Instructs an AXI RX DMA to read the RX FIFO. If the DMA is set to perform burst reads, the threshold must ensure that there is sufficient data in the FIFO to satisfy the burst.	R/W	0
REQW6	20	TX DREQ Threshold Level. A TX DREQ will be generated when the TX FIFO falls below this threshold level. Instructs an AXI TX DMA to write more data to the TX FIFO.	R/W	0

SMI Device Read Settings Register The SMI Device Read Settings register is used to configure the timings and bus configuration used for reads of an external device. 4 Device settings registers are available, allowing 3 different configurations to be specified. The read device is selected in the Address register.

Field	Width	Description	Access	Reset
RWLEN	4	Read Transfer Width. 00 = 8-bit ; 01 = 16-bit ; 10 = 18-bit; 11 = 9-bit.	R/W	00
RSETUP	20	Duration between chip select being asserted and read strobe going active. Measured in SMI bus clock cycles [1-64]	R/W	0
MODE68	1	System-x Bus Operation. Specify the operation mode of the external bus. 0 = System-80 ; 1 = System-68	R/W	0
FSETUP	1	Setup Time Multiplicity. Specify wether to apply setup time to each transfer. 0 = Always apply setup time ; 1 = Setup time only applied to the first transfer after de-assertion of chip select.	R/W	0
RHOLD	20	Read Strobe Hold Duration. Duration between read strobe going inactive and chip select de-asserting. Measure in SMI bus clock cycles [1-64]	R/W	0

Field	Description	Access	Reset
RPACE	Pace Time Multiplicity. Specify whether to apply pace time to each transfer. 0 = Only apply to access through the same device settings ; 1 = Apply to next access regardless of device settings used	R/W	0
RPACE	Duration between chip select de-asserting and any other transfer being allowed on the bus. Measured in SMI bus clock cycles [1-128]	R/W	00
RDRREQ	External Read DREQ. The top 2 bits of the SMI data bus are used as DREQ (SD[16]) and DREQ_ACK (SD[17]) to pace the reads. Must be used in conjunction with DMAP in SMI_CS. 0 = Don't use external DMA requests, reads always happen ; 1 = Use external DMA request to pace device reads, only happen for each DREQ, DREQ_ACK cycle.	R/W	0
RSTROBE	Read Strobe Duration. Duration that the read strobe should be active. Measured in SMI bus clock cycles [1-128]	R/W	0C

SMI Device Write Settings Register The SMI Device Write Settings register is used to configure the timings and bus configuration used for writes to an external device. 4 Device settings registers are available, allowing 3 different configurations to be specified. The write device is selected in the Address register.

Field	Description	Access	Reset
WWID	Write Transfer Width. 00 = 8-bit ; 01 = 16-bit ; 10 = 18-bit; 11 = 9-bit	R/W	00
WSE	Duration between chip select being asserted and write strobe going active. Measured in SMI bus clock cycles [1-64]	R/W	01
WFORMAT	Write Pixel Format. Specifies the packing method used when data is packed. 0 = 16-bit RGB565 ; 1 = 32-bit RGBX8888	R/W	0
WSWAP	Swap Modes. Specifies if word read bit order is swapped. 0 = Pixel data bits not swapped ; 1 = Pixel data bits swapped	R/W	0
WHOLD	Write Strobe Hold Duration. Duration between write strobe going inactive and chip select de-asserting. Measured in SMI bus clock cycles [1-64]	R/W	01
RPACE	Pace Time Multiplicity. Specify whether to apply pace time to each transfer. 0 = Only apply to access through the same device settings ; 1 = Apply to next access regardless of device settings used	R/W	0
WPACE	Duration between chip select de-asserting and any other transfer being allowed on the bus. Measured in SMI bus clock cycles [1-128]	R/W	8

Field	Width	Description	Access	Reset
WDREQ	2	External Write DREQ. The top 2 bits of the SMI data bus are used as DREQ (SD[16]) and DREQ_ACK (SD[17]) to pace the writes. Must be used in conjunction with DMAP in SMI_CS. 0 = Don't use external DMA request, writes always happen ; 1 = Use external DMA request to pace device writes, only happen for each DREQ, DREQ_ACK cycle.	R/W	0
WSTBROW	8	Write Strobe Duration. Duration that the write strobe should be active. Measured in SMI bus clock cycles [1-128]	R/W	0x0C

SMI Direct Control and Status Register The SMI Direct Control and Status register is used to control direct mode transfers. A direct write will be scheduled even if a normal data transfer is taking place.

Field	Width	Description	Access	Reset
__X	31:4			
WRITE	1	Transfer Direction. 0 = Transfers will read from external devices ; 1 = Transfers will read from external devices	R/W	0
DONE	1	Direct Transfer Complete. 0 = No meaning ; 1 = A transfer has finished. Writing 1 to this bit will clear this flag	R/W	0
START	1	Start Transfer. 0 = No effect ; 1 = Writing a 1 to this bit will start a transfer if one is not already taking place	W	0
ENABLE	1	Enable the SMI. This bit is OR'd with the ENABLE bit in the SMI_CS register. If either bit is set then the SMI is enabled. 0 = SMI is disabled and is placed into in a power-saving state with minimal clocking, however the SMI control registers can still be read and written ; 1 = Enable the SMI	R/W	0

SMI Direct Mode Address Register The Direct Mode Address register is used to specify the address presented on the external pins when a Direct Mode transfer takes place. The device settings used is also selected. The register should be written with the address and device settings before the direct mode transfer is placed.

Field	Width	Description	Access	Reset
__X	31:10			
DEVSEL	4	Device Settings Selection. 00 = Device settings 0 ; 01 = Device settings 1 ; 10 = Device settings 2 ; 11 = Device settings 3.	R/W	0x0
__X	7:6			
ADDR	8	Address to be used for direct mode transfers	R/W	0x00

SMI Direct Mode Data Register The Direct Mode Data register is used to read/write the data for a Direct Mode transfer. Write data must be written here before the Direct mode write is started. Read data is valid once the read has completed as indicated by the DONE bit in the SMI_DCS register.

Field	Width	Description	Access	Reset
DATA	31:0	Reading returns data that has been read from external devices. Writing is used to provide the data that should be written to external devices	R/W	0

SMI FIFO Debug Register The SMI FIFO Debug register is used to show FIFO status for debug use

Field	Width	Description	Access	Reset
FLVL	31:14	FIFO High Count Level. Displays the maximum FCNT level reached during the last set of transfers. Reset at the start of each new transfer.	R/W	0
FCNT	7:0	FIFO Count. Displays the FIFO fill level of the currently active FIFO. if doing a write transfer then it will indicate the TX FIFO, else the RX FIFO	R/W	0